Group 3

Air Quality and Pollution

Allotted dataset: Air Quality and Pollution

Dataset source: UCI Air Quality - open source and documentation

Read the source description and variable definitions to understand the allotted dataset before beginning your work.

Project objective

Investigate how air pollution changes over time and how it relates to weather conditions and sensor measurements. Develop a model that predicts the next hour's carbon monoxide concentration, recorded as CO(GT), using information available up to the current hour. Explain the usefulness and limitations of your predictions for air-quality monitoring.

Required tasks

1. Data preprocessing

2. Exploratory data analysis

3. Predictive model building and evaluation

4. Hyperparameter tuning

Choose and justify your approach

Your group must decide the preprocessing methods, statistical techniques, models, evaluation measures and hyperparameter-tuning strategy. Justify each choice in relation to the data and the project objective, and interpret the results.

All work must be carried out in Python only.

Submission

Submit the Python notebook or scripts, the cleaned dataset and a report presenting your approach, findings, model results, limitations and each member's contribution. Complete the project within 21 days.